

Abhinav Karthik

AI Engineer · Boston, MA

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SUMMARY	Master's in AI from Northeastern (4.0), computer engineering from UMass Amherst. I've shipped a long-term memory system for a commercial agent, an optimizer that schedules a cardiovascular service line, and a Battleship engine that beats a record that stood since 2011.	
EXPERIENCE	Jun 2026 – Now Boston, MA	AI Engineer · State Street via Xoriant Building LLM-driven contract analysis and extraction workflows on Evisort for a global custodian bank. Retrieval, extraction and evaluation work so legal teams can trust what the model says about a document.
	Jan 2026 – Now Portland, ME	AI Research Engineer · MaineHealth · Roux Institute Built a mixed-integer linear programming system that generates annual schedules for ~60 physicians across 5 subspecialty groups and 46 services. Designed a cascading multi-stage solver on Google OR-Tools CBC with a registry-based constraint engine supporting 14 configurable hard and soft constraint types. Block variable compression, per-group skeleton decomposition and upstream-aware constraint propagation: 91% happiness score across 8,500+ weekly assignments in under 4 minutes. Shipped it as a Flask web app with drag-and-drop pre-filled uploads, live solve progress and Excel export for clinical deployment.
	Mar 2026 – Now Boston, MA	Lead Engineer · 7D Connect · Northeastern Architected a Next.js 16 monorepo where one web app serves the admin UI and the mobile REST API over a single Prisma/PostgreSQL schema. Native SwiftUI and Kotlin/Compose driver apps: route start/finish, GPS-stamped attendance, inspection checklists, incident reports and mileage. Role model spanning super admin, vendor admin, district and driver, with vendor impersonation for support and nightly backups.
	Jun 2025 – Sep 2025 Boston, MA · Hybrid	AI Engineer · EasyBee AI Architected a persistent long-term memory system for the primary commercial agent, lifting retention by 15%. Cut agent latency by ~2 seconds through empirical evaluation and context engineering. Led an agentic coding team that delivered an admin dashboard with deep-research capabilities in under two weeks, plus an automated RAG knowledge-base pipeline with industry-specific chunking metadata. Built an MCP-based extension for omni-channel SMS and WhatsApp messaging via Twilio while preserving context and memory. Drove an AI-first initiative automating repeatable work across business and technical teams, averaging a 3.5× productivity gain.
	Jan 2021 – May 2024 Bangalore, India	Founder · BucketListEr Conceptualised, built and released the app; connected users with like-minded people and partners offering discounts. Hired and ran two intern teams (investor search, marketing) at near-zero cost.

Jun 2024 – Jul 2024		IT Intern · SFJ Business Solutions	India	Wrote the sections of a Python, Pandas and MATLAB testing tool that measured algorithmic speed and surfaced sub-optimal paths.
RESEARCH	2026 under review	GA-tuned ensemble agent for Battleship		
		Four Battleship strategies, one heatmap each, blended by weights a genetic algorithm learned. Averages 40.5 moves to win and beats the 2011 benchmark. with Jonathan Mwaura		
RESEARCH	2026 under review	Adaptive generational ↔ steady-state switching for GAs		
		A genetic algorithm that starts generational, watches its own elite centroid, and switches itself to steady-state the moment exploration stalls. No hand-picked switch point. with Jonathan Mwaura		
SELECTED PROJECTS	2026	7D Connect — School-bus operations platform: web admin, REST backend, native iOS and Android driver apps.		
	2026 – now	Legal intelligence at State Street — What I do now: LLM workflows for contract intelligence on Evisort.		
	2026	Luna OS — An LLM-native operating system shell built with Tauri and React.		
	2026	Genome Explorer — A 3D DNA helix you can orbit, where the pathogenic variant glows at 60 bpm.		
EDUCATION	Sep 2024 – May 2026	Northeastern University — M.S. Artificial Intelligence · GPA 4.0 / 4.0		
	Jun 2020 – May 2024	University of Massachusetts Amherst — B.S. Computer Engineering · Phi Delta Theta, founding member & VP		
SKILLS	AGENTS & LLMS	Claude Code · Codex · Cursor · MCP · LangGraph · LangChain · RAG · Vector DBs · Prompt engineering · Fine-tuning · Evals · Voiceflow		
	ML & RL	PyTorch · TensorFlow · Keras · scikit-learn · OpenCV · PPO / MARL · CNNs · Transformers · Pandas · NumPy		
	OPTIMISATION	Genetic algorithms · OR-Tools · MILP · Monte Carlo · Bayesian optimisation · Constraint programming · Information theory		
	LANGUAGES	Python · TypeScript · Java · C++ · C# · Embedded C · MATLAB · Swift · Kotlin		
	WEB, MOBILE & INFRA	Next.js · React · Tauri · SwiftUI · Jetpack Compose · Flask / FastAPI · PostgreSQL · Prisma · Supabase · Pinecone · Docker · AWS · Vercel · Twilio		
Generative AI with Large Language Models — DeepLearning.AI × AWS, 2024				
Generative AI: Prompt Engineering Basics — IBM, 2024				
Python for Data Science, AI & Development — IBM, 2024				